

Review

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Common Units and Their Rates

Decimals, Fractions, Percentages

Rectangular Prisms and Cubes

Circles

Venn Diagram

Ratio

Cylinders and Cones

Permutation and Combination

Pigeonhole Principle

Common Units and Their Rates

Common Units and Their Rates

1. Length Units:

kilometer (km) $\xrightarrow{1000}$ meter (m) $\xrightarrow{10}$ decimeter (dm) $\xrightarrow{10}$ centimeter (cm) $\xrightarrow{10}$ millimeter (mm)

2. Area Units:

Square kilometer (km²) $\xrightarrow{100}$ hectare (ha) $\xrightarrow{10000}$ Square meter (m²) $\xrightarrow{100}$ Square decimeter (dm²) $\xrightarrow{100}$ Square centimeter (cm²)

3. Volume Units:

Cubic meter (m³) $\xrightarrow{1000}$ Cubic decimeter (dm³) $\xrightarrow{1000}$ Cubic centimeter (cm³)

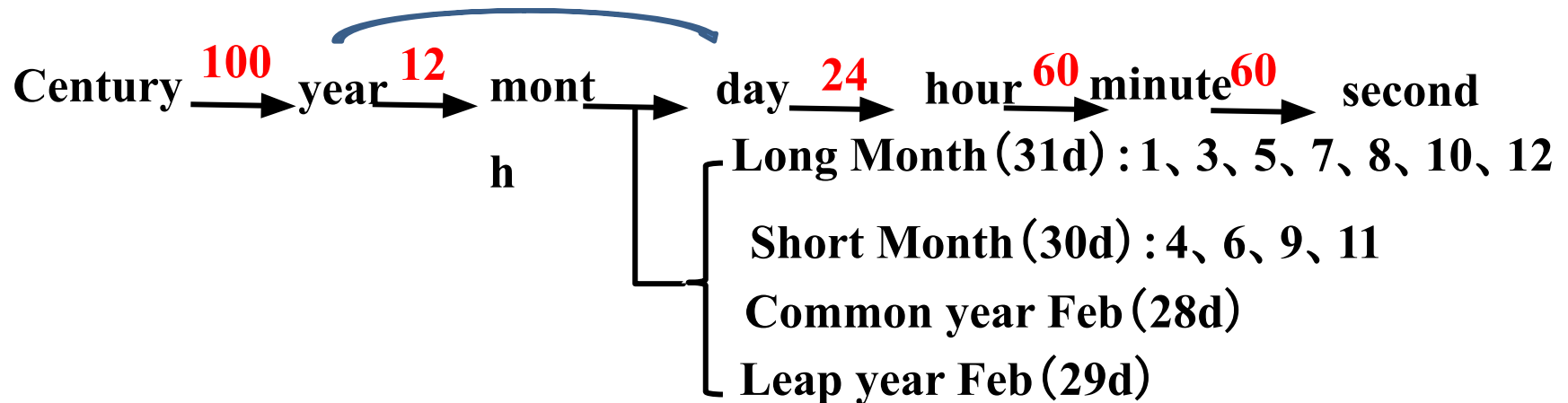
4. Capacity Units:

liter(l) $\xrightarrow{1000}$ milliliter(ml)

Common year 365 days

5. Time Units:

Leap year 366 days



6. Mass Units:

Ton (t) $\xrightarrow{1000}$ Kilogram $\xrightarrow{1000}$ gram (g)

(kg)

7. Currency Units:

dollar $\xrightarrow{10}$ dim $\xrightarrow{10}$ cent

e

Fill in the blanks

$$0.4\text{m}=(\text{ }40\text{ })\text{cm}$$

$$725\text{mm}=(\text{ }7.25\text{ })\text{dm}$$

$$3.2\text{m}^2=(\text{ }320\text{ })\text{dm}^2$$

$$5\text{dm}^2=(\text{ }500\text{ })\text{cm}^2$$

$$7500\text{mL}=(\text{ }7.5\text{ })\text{L}$$

$$6.2\text{dm}^2=(\text{ }620\text{ })\text{cm}^2$$

$$0.24\text{km}^2=(\text{ }240000\text{ })\text{m}^2$$

$$4160\text{cm}^2=(\text{ }41.6\text{ })\text{dm}^2$$

$$2.8\text{L}=(\text{ }2800\text{ })\text{mL}$$

$$8.75\text{m}^3=(\text{ }8750\text{ })\text{dm}^3$$

$$64\text{cm}^3=(\text{ }0.064\text{ })\text{m}^3$$

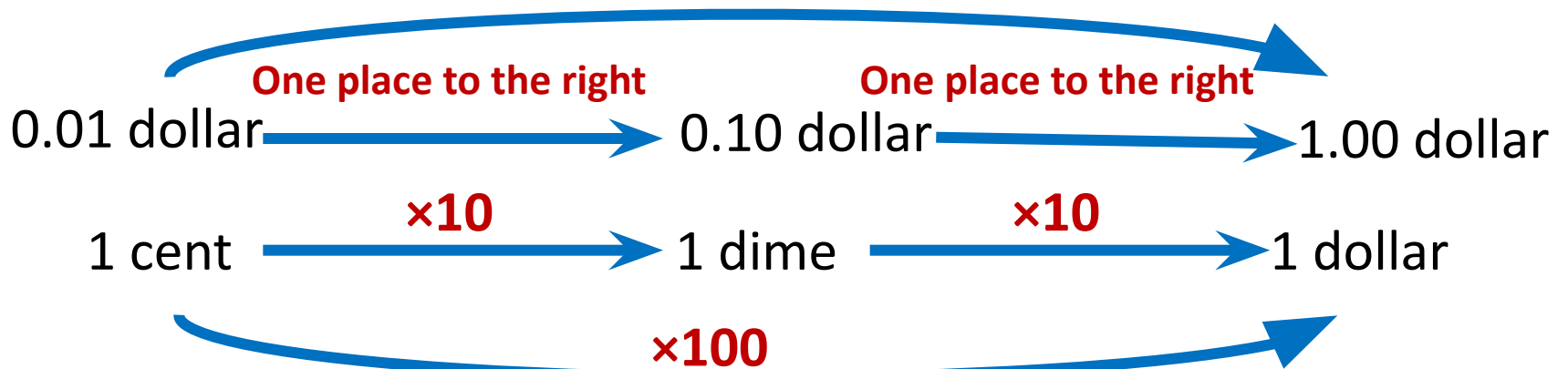
$$320\text{mL}=(\text{ }320\text{ })\text{cm}^3$$

Decimals, Fractions, Percentages



What happens to the size of a decimal when the decimal point is moved to the right ?

Move the decimal point two places to the right.

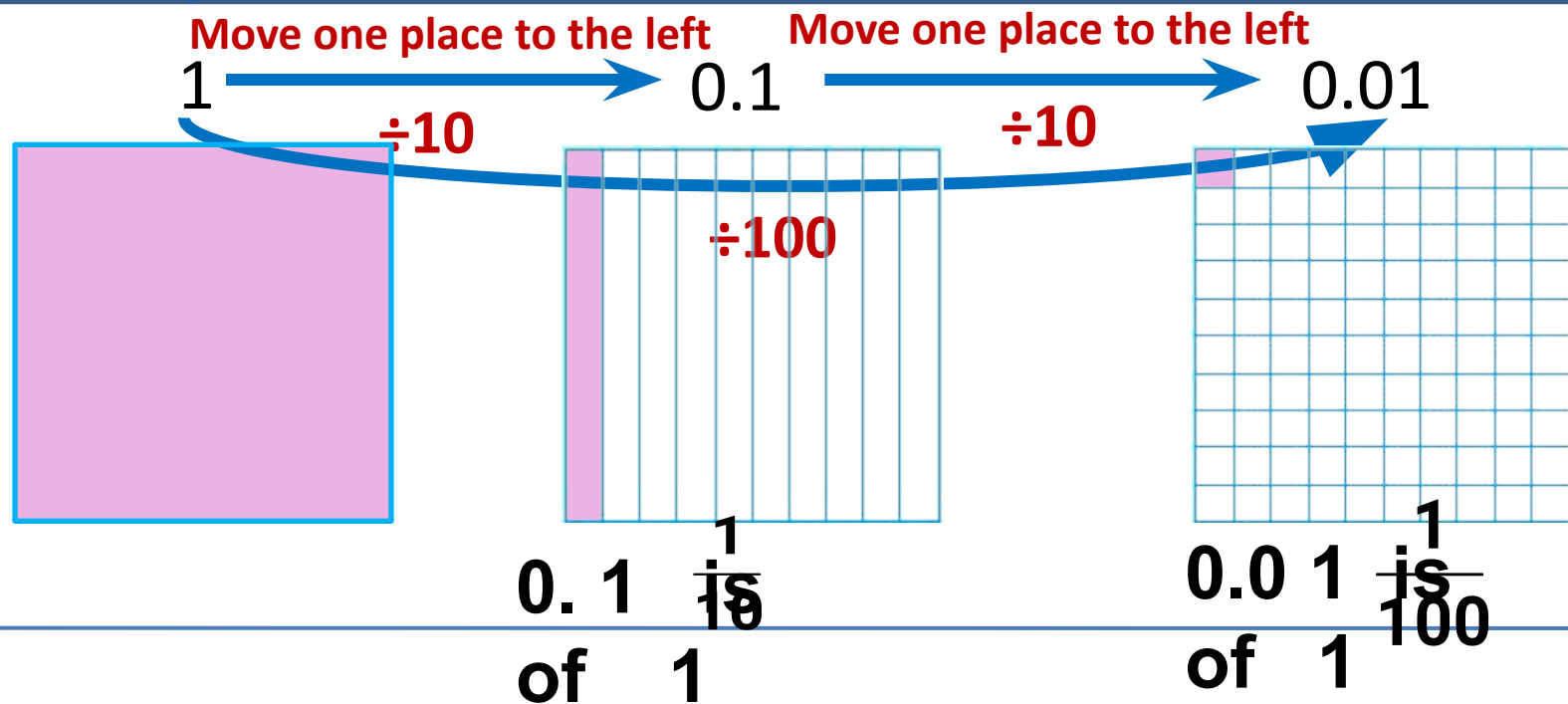




What happens to the size of a decimal when the decimal point is moved to the left ?

When the decimal point of a number is moved one place to the left, the resulting number is $\frac{1}{10}$ of the original number

When moved two places, it is $\frac{1}{100}$ of the original number



When the decimal point of a number is moved **one place to the right**, the resulting number is **10 times** the original number .

When the decimal point of a number is moved **two places to the right**, the resulting number is **100 times** the original number .

When the decimal point of a number is moved **one place to the left**, the resulting number is $\frac{1}{10}$ the original number .

When the decimal point of a number is moved **two places to the left**, the resulting number is $\frac{1}{100}$ the original number .

Practice

To determine the number of decimal places in the product, simply count the total number of decimal places in the two multipliers.

Place the decimal point in the following products.

$$4.7 \times 1.5 = 70\dot{5}$$

$$9.32 \times 1.4 = 1304\dot{8}$$

$$1.87 \times 2.3 = 430\dot{1}$$

$$1.6 \times 32 = 512\dot{.}$$

Explore



Wrapping a gift box requires 0.8 meters of wrapping paper and 2.4 meters of ribbon.

Wrapping paper: 2.6 dollars/ m
Ribbon: 0.85 dollars/m

How much does it cost to buy the ribbon?

$$2.4 \times 0.85 = 2.04 \text{ dollars}$$

2.4 One decimal place

$$\begin{array}{r}
 \times 0.85 \\
 \hline
 120 \\
 192 \\
 \hline
 2040
 \end{array}$$

Two decimal places

Three decimal places

Practice

Place the decimal points in the products of the following problems.

$$2.4 \times 2.8 = 672$$

$$4.23 \times 0.8 = 3384$$

$$40.2 \times 11 = 4422$$

$$0.15 \times 1.7 = 255$$

$$0.4 \times 38.2 = 1528$$

$$0.56 \times 0.67 = 03752$$

The calculation method for decimal multiplication

Ignore the Decimal Point: First, multiply the numbers as if they were whole numbers, ignoring the decimal points.

Multiply as Usual: Perform the multiplication just as you would with whole numbers.

Count the Decimal Places: After multiplying, count the total number of decimal places in the numbers you multiplied (i.e., the number of digits to the right of the decimal points in both numbers).

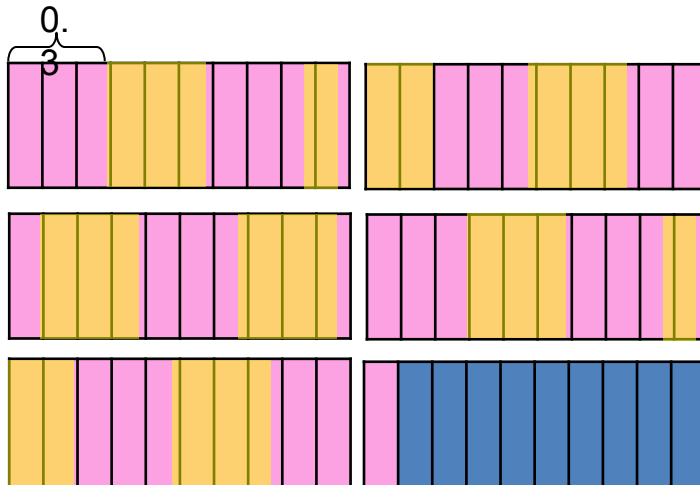
Place the Decimal Point: In the product, place the decimal point so that the number of decimal places in the result equals the total number of decimal places counted in step 3. If there are not enough digits, add zeros to the left of the product. If the product has trailing zeros, place the decimal point first and then remove the trailing zeros.

Explore



Domestic calls cost 0.3 dollars per minute, and the total cost was 5.1 dollars.

How many minutes of calls were there in total?



$$5.1 \div 0.3 = 17$$

$$\begin{array}{r} 17 \\ \cancel{0.3} \overline{) 5.1} \\ \underline{3} \\ 21 \\ \underline{21} \\ 0 \end{array}$$

Practice

$$0.544 \div 0.16 = 3.4 \quad 25.6 \div 0.032 = 800$$

$$\begin{array}{r}
 3.4 \\
 0.16 \overline{) 0.544} \\
 \underline{48} \\
 64 \\
 \underline{64} \\
 0
 \end{array}$$

$$\begin{array}{r}
 800 \\
 0.032 \overline{) 25.600} \\
 \underline{256} \\
 0
 \end{array}$$

To convert the divisor 0.032 into an integer, how many places should the decimal point be moved to the right?

Conclusion

When calculating division where the divisor is a decimal, based on the rule that 'the quotient remains unchanged when both the dividend and divisor are multiplied by the same factor (except 0),'

First remove the decimal point from the divisor to make it an integer.

Then, check how many decimal places the original divisor had, and move the decimal point in the dividend to the right by the same number of places. If there are not enough digits, add zeros to make up the difference.

Finally, perform the calculation according to the method for dividing decimals by an integer divisor.

Calculate the following questions and indicate which of the quotients are repeating decimals.

$$1 \div 2 = 0.5$$

$$1 \div 3 \approx 0.333\ldots$$

$$1 \div 5 = 0.2$$

$$1 \div 7 \approx 0.142857142857\ldots$$

A decimal where, from a certain digit onward, one or several digits repeat endlessly is called a **repeating decimal**. For example, $0.333\ldots$ and $0.142857142857\ldots$ are repeating decimals.



$$15 \div 16 = 0.9375$$

Terminating decimal

$$28 \div 18 = 1.555\cdots$$

Non-terminating decimal

$$78.6 \div 11 = 7.14545\cdots$$

Non-terminating decimal

A decimal with a finite number of digits in the decimal part is called a **terminating decimal**.

A decimal with an infinite number of digits in the decimal part is called a **non-terminating decimal**.



The sequence of digits that repeatedly appears in the decimal part of a repeating decimal is called the **repeating block of the decimal**.



Repeating block of $5.333 \dots$ is 3

Repeating block of $2.08181 \dots$ is 81

Repeating block of $6.9258258 \dots$ is 258

When writing a repeating decimal, you can write only the first repeating block and place a dot above the first and last digits of this block.



5.333 ... 5. $\dot{3}$

2.08181... 2.0 $\dot{8}1$

$\dot{6}$.9258258... 6. $\dot{9}2\dot{5}8$

$$12 \div \frac{4}{5} \div \frac{3}{8}$$

$$= \cancel{12}^1 \times \frac{5}{\cancel{4}_1} \times \frac{8}{\cancel{3}_1}$$

$$= 40$$

$$\frac{3}{8} \div \frac{1}{10} \times \frac{5}{7}$$

$$= \frac{\cancel{3}_4}{\cancel{8}_4} \times \frac{\cancel{10}^5}{1} \times \frac{5}{7}$$

$$= \frac{75}{28}$$

$$\frac{4}{7} \boxtimes \frac{5}{8} \div \frac{5}{14}$$

$$= \frac{\cancel{4}_1}{7} \times \left(\frac{\cancel{5}_4}{\cancel{8}_4} \times \frac{\cancel{14}^7}{\cancel{5}_1} \right)$$

$$= \frac{\cancel{4}_1}{7} \times \frac{\cancel{7}_1}{\cancel{4}_1}$$

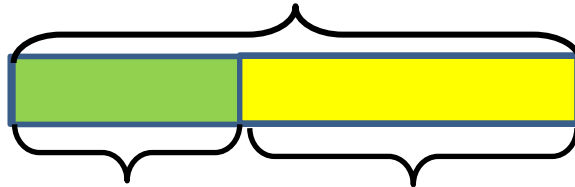
$$= 1$$

The order of operations for mixed operations with fractions is the same as that for mixed operations with whole numbers.



There are 40 students, of which the number of girls accounts for $\frac{2}{5}$ of the total class. How many boys are there?

40 people



boys?

Method 1: First, calculate how many girls there are

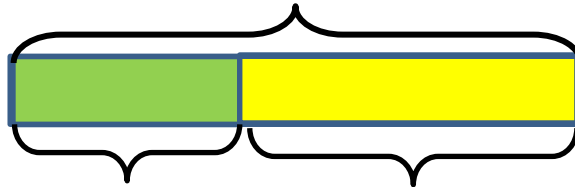
$$= 40 - 16$$

$$= 24(\text{boys})$$



There are 40 students, of which the number of girls accounts for $\frac{2}{5}$ of the total class. How many boys are there?

40 people



boys?

Method 2: First, calculate what fraction of the total class the boys make up.

= 24(boys)

Conclusion

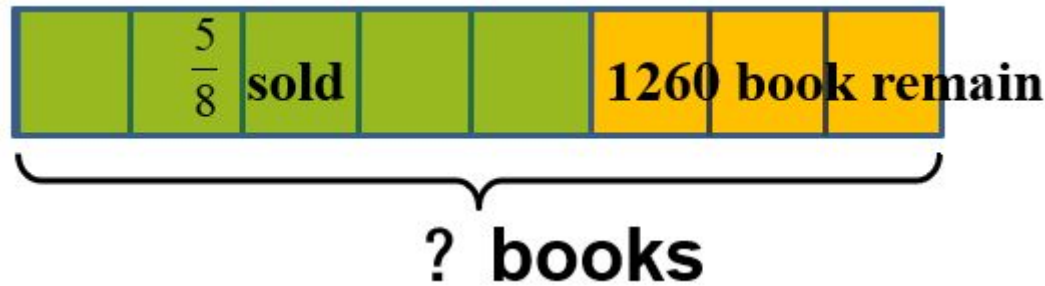
To solve the problem of finding the quantity of the other part when you know what fraction one part represents of the total quantity

First, determine the quantity of one part, and then find the result by using

Quantity of other part = Total quantity – Quantity of known part

First, determine what fraction of the total quantity the other part represents, and then find the result by using

Quantity of other part = Total quantity × Fraction of total quantity the other part represents



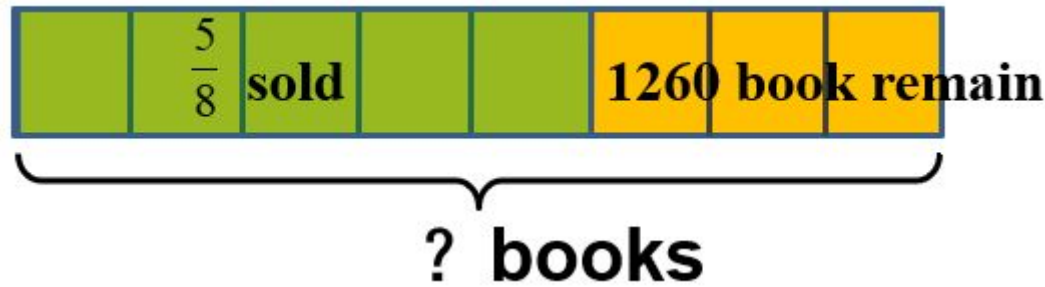
Total number of books - Number of books sold = Number of books remaining.

Method 1

Let x represent the total number of literature books in this shipment.

$$\begin{aligned}
 x - \frac{5}{8}x &= 1260 \\
 \frac{3}{8}x &= 1260 \\
 x &= 3360
 \end{aligned}$$

Answer: There are 3,360 literature books in this shipment.



Total number of books $\times (1 - \frac{5}{8}) = \text{number of books remaining}$

Method 2

Let x represent the total number of literature books in this shipment.

$$\begin{aligned} (1 - \frac{5}{8})x &= 1260 \\ \frac{3}{8}x &= 1260 \\ x &= 3360 \end{aligned}$$

Answer: There are 3,360 literature books in this shipment.

Conclusion

The method for solving problems where "a fraction of the total quantity and the amount of the other part are given, and the task is to find the total quantity" is:

(1)

Let the total quantity be x . Then, the equation can be set up as follows:

$$x - (\text{fraction of the total quantity} \times x) = \text{amount of the other part}$$

(2)

Let the total quantity be x . Then, the equation can be set up as:

$$(1 - \text{fraction of the total quantity}) \times x = \text{amount of the other part}$$

Convert the following numbers into percentages.

$$0.24 = 24\%$$

$$1.76 = 176\%$$

$$0.05 = 5\%$$

$$\frac{3}{4} = 75\%$$

$$\frac{5}{8} = 62.5\%$$

$$\frac{1}{6} \approx 16.7\%$$

Explain how decimals and fractions are converted into percentages?

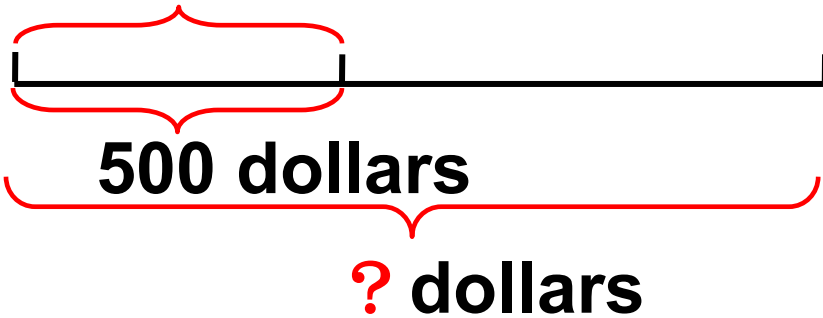


Conclusion

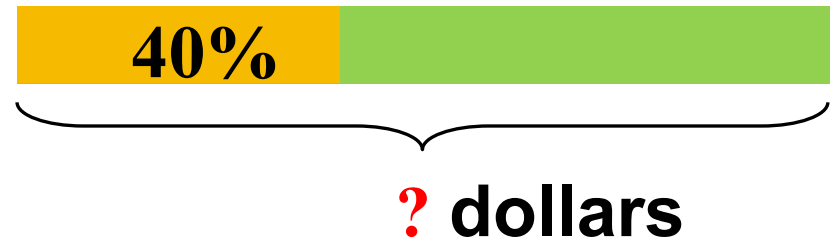
- To convert a decimal into a percentage: first, **move the decimal point two places to the right**, then **add a percent sign at the end**.
 - To convert a fraction into a percentage: you can either rewrite the **fraction with a denominator of 100** and then convert it into a percentage, or first **convert the fraction into a decimal** (if it cannot be divided evenly, typically **round to three decimal places**), and then convert it into a percentage.
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● Can you find the equivalent relationship?

40%



500 dollars



Total expenses $\times$ 40% = Amount spent on food

List the equation, then solve the problem

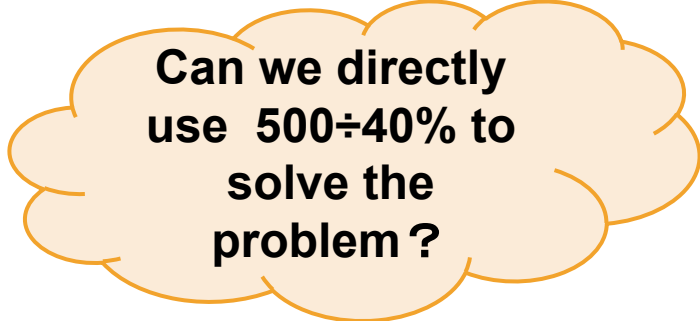
$$\text{Total expenses} \times 40\% = \text{Amount spent on food}$$

Let the total expenditure of Jerry's family this month be x dollars

$$40\%x = 500$$

$$x = 500 \div 40\%$$

$$x = 1250$$

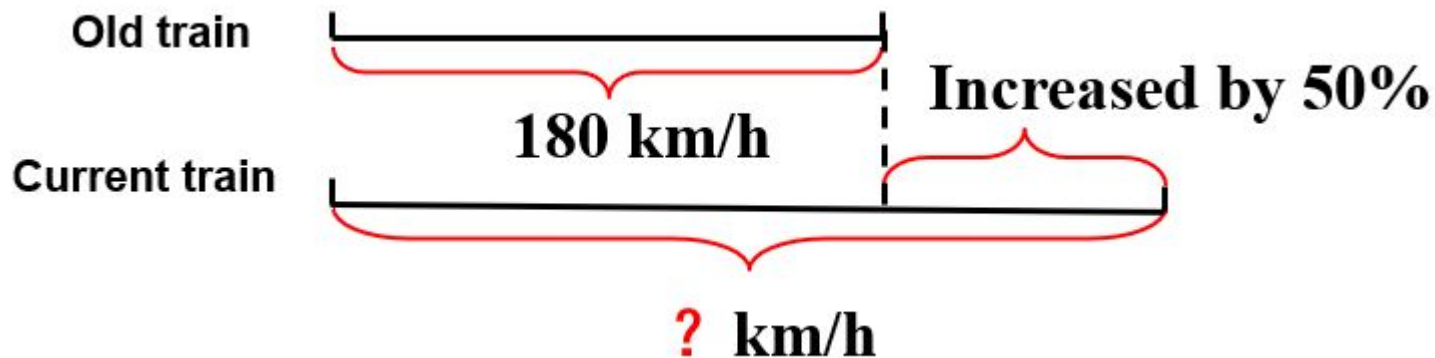


Can we directly
use $500 \div 40\%$ to
solve the
problem?

Answer : The total expenditure of Jerry's family this month is 1250 dollars.

Conclusion

To solve the problem "Given a certain percentage of a number, find the original number," you can identify the equivalent relationship based on the keywords, **set up an equation, and solve it**; alternatively, you can **directly use division to calculate the answer**.

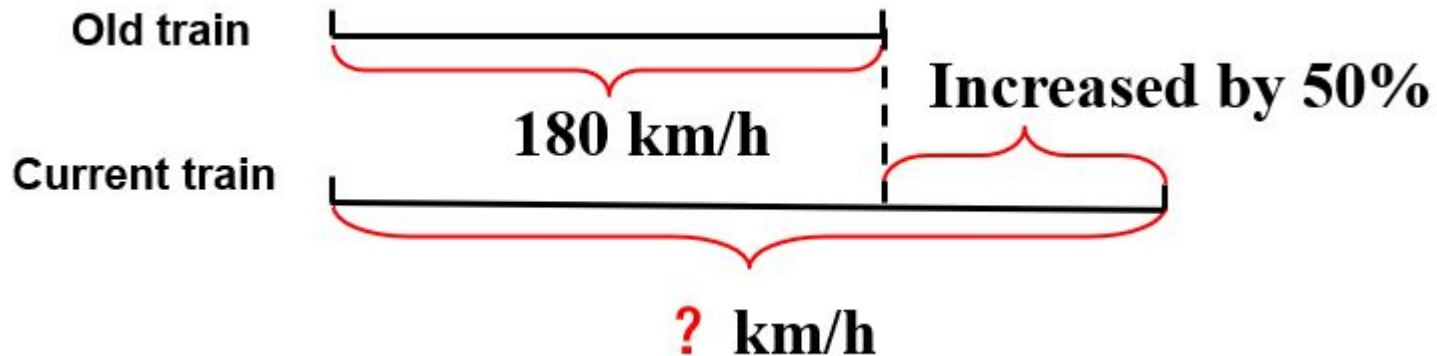


First, calculate how many more kilometers are traveled per hour compared to the original speed.

$$180 \times 50\% = 90 \text{ (km/h)}$$

$$180 + 90 = 270 \text{ (km/h)}$$

Answer: The current high-speed train travels 270 kilometers per hour.



First, calculate what fraction the current speed is of the original speed.

$$\begin{aligned} &180 \times (1 + 50\%) \\ &= 180 \times 1.5 \\ &= 270 \text{ (km)} \end{aligned}$$

Answer: The current high-speed train travels 270 kilometers per hour.

Conclusion

To solve practical problems of "finding a number that is a certain percentage greater than another number," you can use two methods:

(1) $A \text{ number} + (a \text{ number} \times \text{the percentage increase})$

(2) $A \text{ number} \times (1 + \text{the percentage increase})$

Explore

Savings mainly come in two types: **fixed-term** and **current**.

Principal: The money deposited into the bank.

Interest: The additional money paid by the bank, apart from the principal, when withdrawing.

Interest Rate: The percentage of the principal that the interest represents. The annual interest rate is the rate calculated per year.

$$\text{Interest} = \text{principal} \times \text{interest rate} \times \text{time}$$



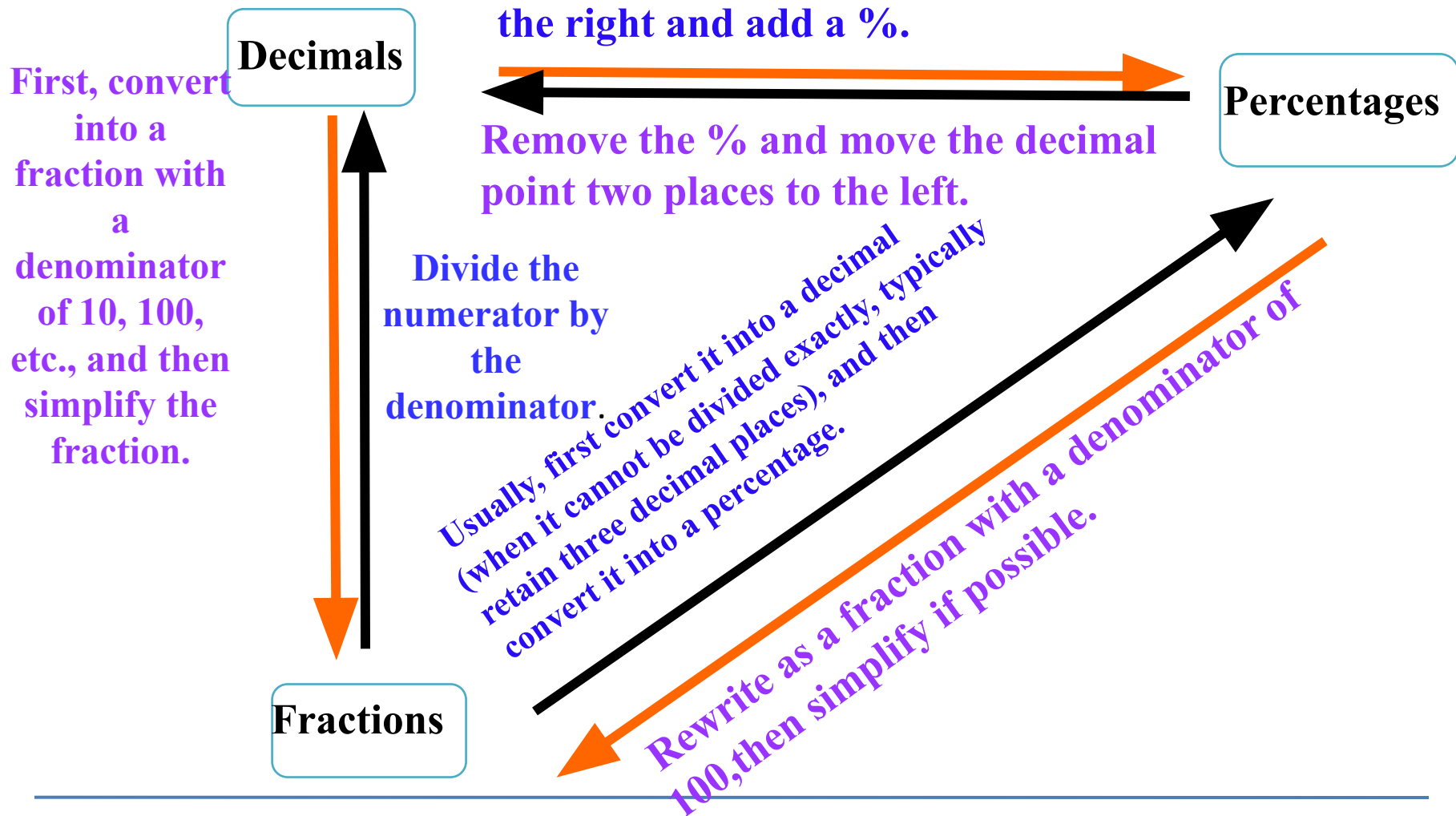
If \$300 is deposited for three years, how much interest will there be at maturity?

$$\begin{aligned} & 300 \times 3.33\% \times 3 \\ &= 300 \times \frac{3.33}{100} \times 3 \\ &= 9.99 \times 3 \\ &= 29.97 \text{ (dollars)} \end{aligned}$$

fixed-term deposit	Annual rate/%
One year	2.25
Three years	3.33
Five years	3.60

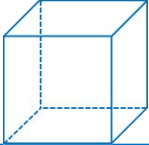
Answer: There will be 29.97 dollars in interest at maturity。

Conversion of Fractions, Decimals, and Percentages



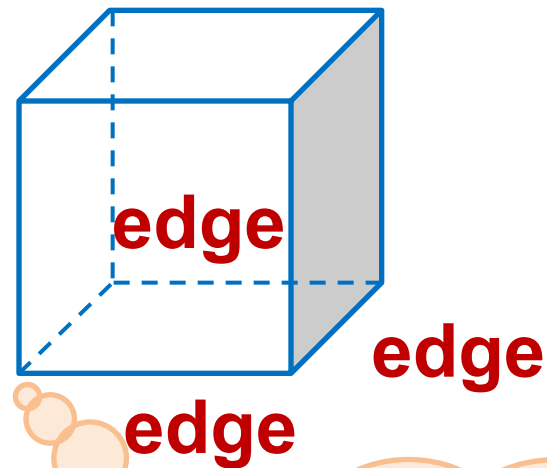
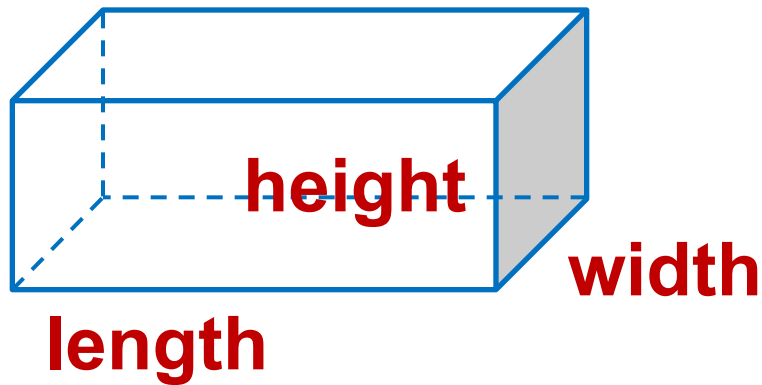
Rectangular Prisms and Cubes

Explore

			
vertex	number	8	8
face	number	6	6
	shape	Square	Rectangle or square
	size	Equal size	Equal size of two opposite faces
edge	number	12	12
	relationship	Equal length	Equal length of opposite edges

Explore

● Square a special type of rectangle, and is a cube a special type of rectangular prism?



A cube can be considered a rectangular prism with equal length, width and height.

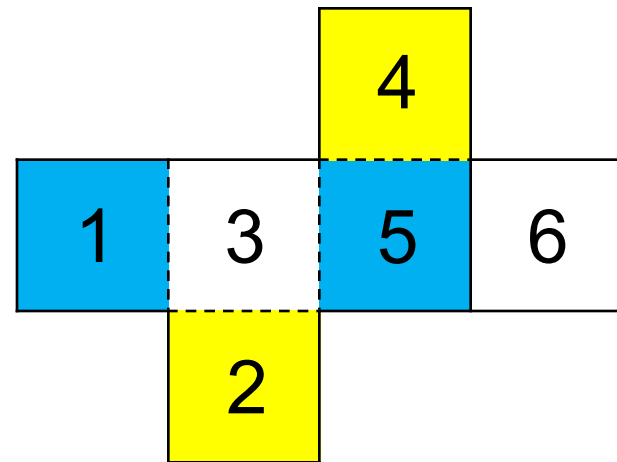
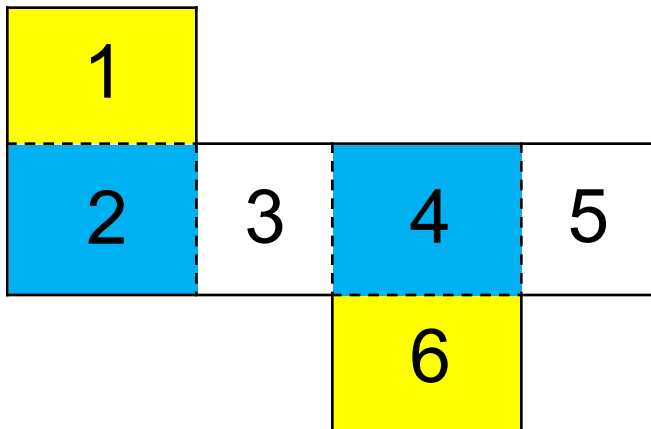
Conclusion

A rectangular prism is formed by 6 rectangles (in special cases, two opposite faces are squares). In a rectangular prism, opposite faces are identical, and opposite edges have equal lengths. The three edges that intersect at a vertex represent the length, width, and height of the rectangular prism.

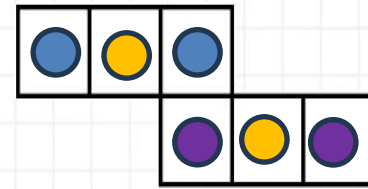
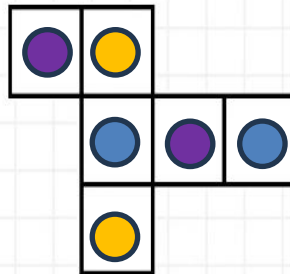
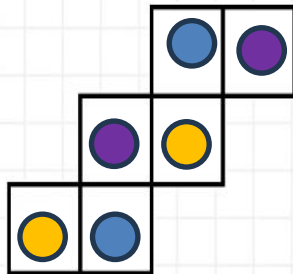
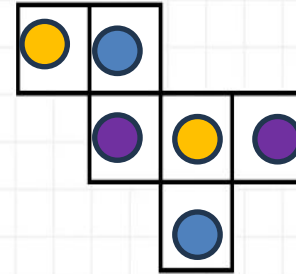
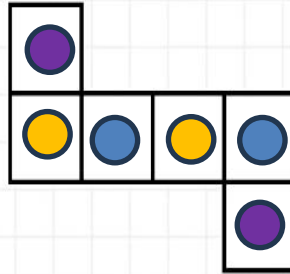
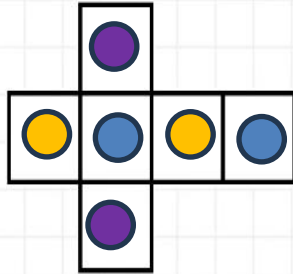
A cube is a three-dimensional shape formed by 6 identical squares. It has 12 edges of equal length and 8 vertices. A cube is a special type of rectangular prism.

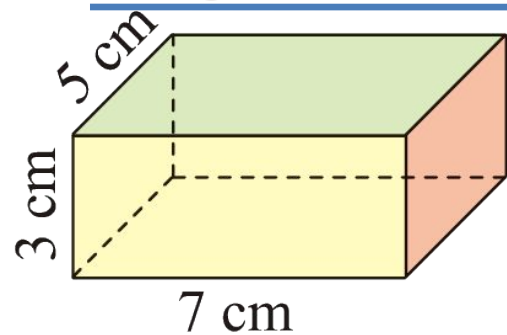
Example

- Below are the net diagrams of a rectangular prism and a cube. Please specify which numbered face is opposite to faces 1, 2, and 3 respectively.

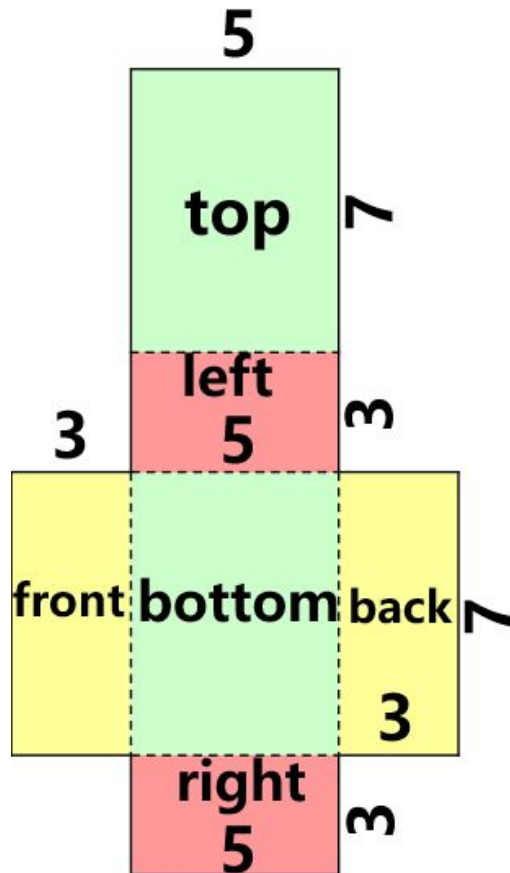


Practice

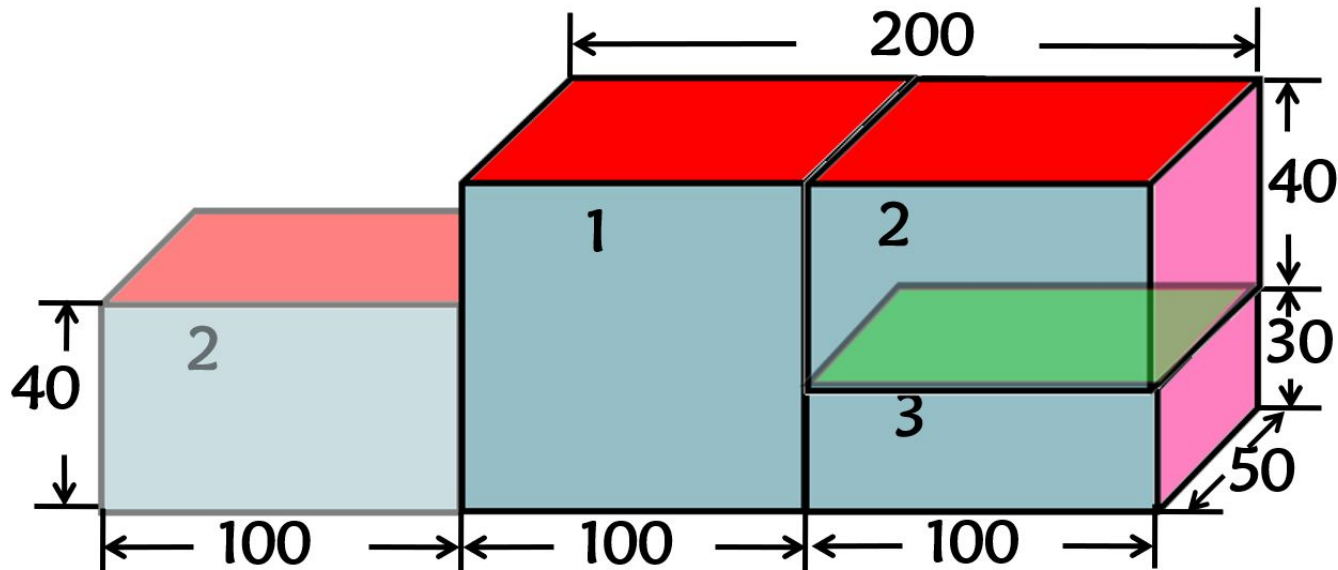




How do you calculate the surface area of this rectangular prism packaging box?



The sum of the areas of the front and back faces.	$7 \times 3 \times 2 = 42 (\text{cm}^2)$
The sum of the areas of the left and right faces.	$3 \times 5 \times 2 = 30 (\text{cm}^2)$
The sum of the areas of the top and bottom faces.	$7 \times 5 \times 2 = 70 (\text{cm}^2)$
Surface area	$42 + 30 + 70 = 142 (\text{cm}^2)$



$$\begin{aligned}
 & 50 \times (30+40) \times 2 + 200 \times (30+40) \times 2 \\
 & + 200 \times 50 + 100 \times 50 \\
 & = 7000 + 28000 + 300 \times 50 \\
 & = 50000(\text{cm}^2)
 \end{aligned}$$

Conclusion

Volume of a rectangular prism = length $\times$ width $\times$ height

$$V = a \times b \times h$$

Volume of a cube = edge $\times$ edge $\times$ edge

$$V = a \times a \times a = a^3$$

The volume of a rectangular prism (cube) = base area $\times$ height

$$V = S \times h = Sh$$

Circles

Characteristics of Circles

A circle has an infinite number of diameters and radii. In the same circle (or congruent circles), all diameters are equal, and all radii are equal.

In the same circle, the length of the diameter is twice the length of the radius, which can be expressed as $d = 2r$ or $r = \frac{d}{2}$

The center of the circle determines its position, while the radius determines its size.

A circle is an axially symmetric shape, and the line containing the diameter serves as an axis of symmetry. A circle has an infinite number of lines of symmetry.

The quotient of the circumference of a circle divided by its diameter is a constant number, which we call pi, represented by the letter π . For calculations, it is usually approximated as **3.14**

$$\pi = 3.141592653\cdots$$

$$\pi \approx 3.14$$

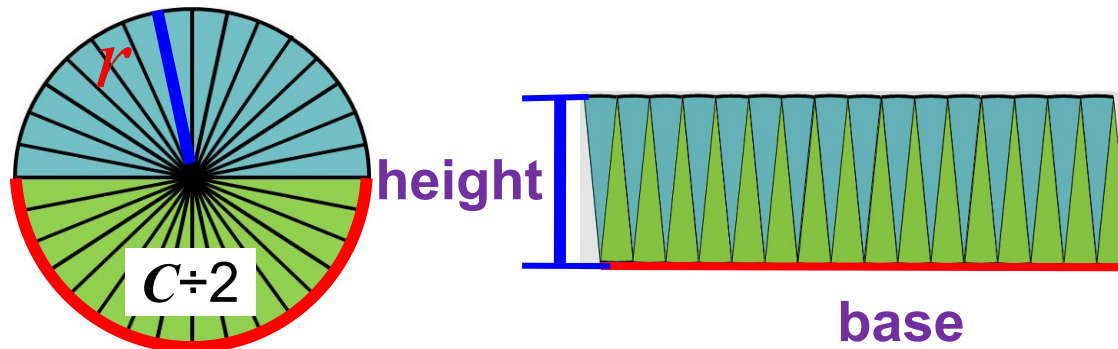
If the radius of a circle is known,
can the circumference be
determined?



Diameter = radius $\times$ 2,

Circumference = radius $\times$ 2 $\times$ pi

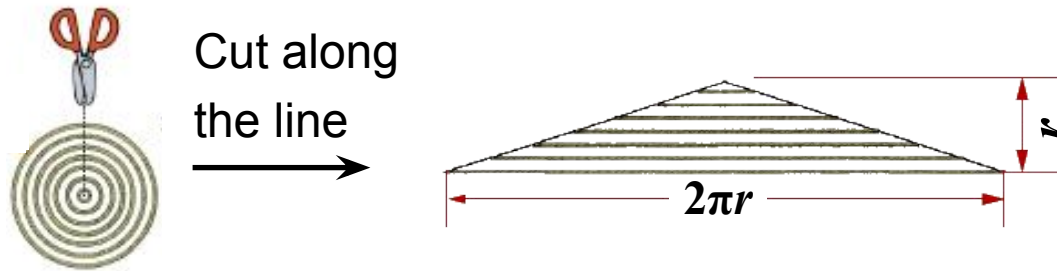
$$C = 2\pi r$$



area of parallelogram = **base** × **height**

$$S = \pi r \times r$$

Area of circle $S = \pi r^2$



The area of the triangle is equivalent to the area of the circle.

The base is equivalent to the (circumference) of the circle, and the height is equivalent to the (radius) of the circle.

$$\text{Area of triangle} = \frac{\text{base} \times \text{height}}{2}$$

$$\text{Area of circle: } S = \frac{2\pi r \cdot r}{2} = \pi r^2$$

Properties of Rotation

Rotation does not change the size or shape of the figure.

Every point on the figure rotates around the center by the same angle and in the same direction.

Any pair of corresponding points forms the rotation angle with the center of rotation.

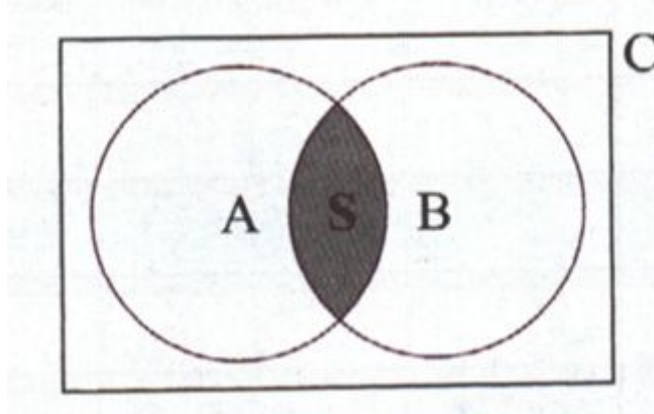
Corresponding points remain equidistant from the center of rotation.

Differences

	Direction of motion	Measurement of motion quantity
Translation	Straight line	Move a certain distance
Rotation	clockwise counterclockwise	Rotate by a certain angle

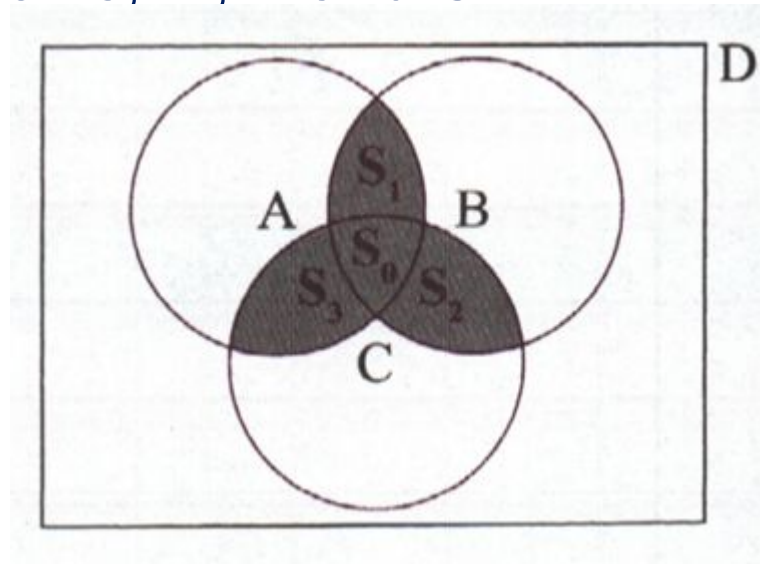
Venn Diagram

The Venn diagram below shows the relationship between two different items, A and B. The shaded area, S, is a region that is common to both A and B.



Total number of items or events, $C=A+B-S$

The Venn diagram below shows the relationship between three different items, A, B and C.



The shaded area, S_1 , is common to both A and B. The shaded area, S_2 , is common to both B and C. The shaded area, S_3 , is common to both A and C. The shaded area, S_0 , is common to both A, B and C.

$$\text{Total number of items or events: } D = A + B + C - S_1 - S_2 - S_3 + S_0$$

Ratio

The relationship between ratios, fractions, and division can be simply represented with letters as follows:

$$a:b = a \div b = \frac{a}{b} \text{ } (b \text{ not equal to } 0)$$

$$1 : 2 = 10 : 20 = \frac{1}{2}$$

$\xrightarrow{\times 10}$ (from 1 to 10) and $\xrightarrow{\times 10}$ (from 2 to 20)

$$4 : 12 = 1 : 3 = \frac{1}{3}$$

$\xrightarrow{\div 4}$ (from 4 to 1) and $\xrightarrow{\div 4}$ (from 12 to 3)

$$14 : 35 = 2 : 5 = \frac{2}{5}$$

$\xrightarrow{\div 7}$ (from 14 to 2) and $\xrightarrow{\div 7}$ (from 35 to 5)



I discovered that when the antecedent and consequent of a ratio are **both multiplied or divided by the same non-zero number**, the value of the ratio remains unchanged.

This is similar to the rule of unchanged quotient and the basic property of fractions that we learned before.



- Fractions can be simplified, and ratios can also be reduced. Can you simplify the following ratios?

$$24:42$$

$$= \frac{24}{42}$$

$$= \frac{4}{7}$$

$$= 4:7$$

$$\frac{2}{5} : \frac{1}{4}$$

$$0.7:0.8$$

First, write the ratio as a fraction (**two numbers divided by each other**), then simplify it, and finally rewrite it as a ratio.



The antecedent and consequent have only a common factor of 1.

Simplest form of a ratio

$$\begin{aligned} & \frac{2}{5} : \frac{1}{4} \\ &= \frac{2}{5} \div \frac{1}{4} \\ &= \frac{2}{5} \times 4 \\ &= \frac{8}{5} \\ &= 8:5 \end{aligned}$$

$$\begin{aligned} & 0.7:0.8 \\ &= 7\div 8 \\ &= 7:8 \end{aligned}$$



Do you have
any other
methods?

$$\begin{aligned} 24:42 &= (24\div 6) : (42\div 6) \\ &= 4:7 \end{aligned}$$

$$\begin{aligned} \frac{2}{5} : \frac{1}{4} &= (\frac{2}{5} \times 20) : (\frac{1}{4} \times 20) \\ &= 8:5 \end{aligned}$$

$$\begin{aligned} 0.7:0.8 &= (0.7\times 10) : (0.8\times 10) \\ &= 7:8 \end{aligned}$$

Practice

A mixed candy is made from milk candy, fruit candy, and crunchy candy in a ratio of 2:4:3. How many kilograms of each type of candy are needed to make 450 kg of this mixed candy?

$$2 + 4 + 3 = 9$$

$$\text{Milk candy: } 450 \times \frac{2}{9} = 100 \text{ (kg)}$$

$$\text{Fruit candy: } 450 \times \frac{4}{9} = 200 \text{ (kg)}$$

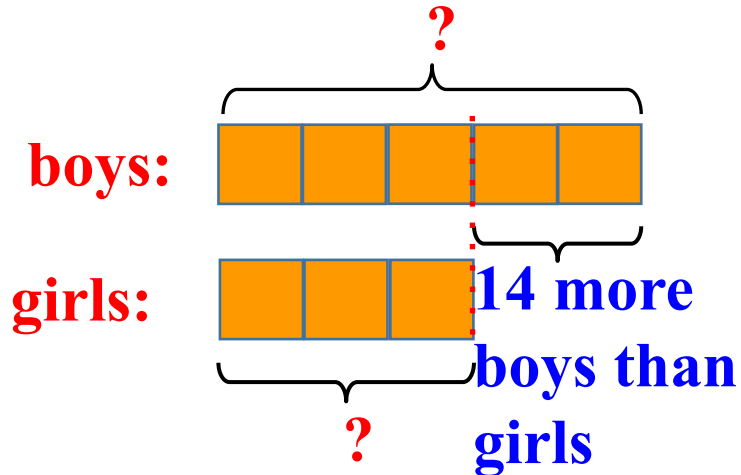
$$\text{Crunchy candy: } 450 \times \frac{3}{9} = 150 \text{ (kg)}$$

Answer: 100 kilograms of milk candy, 200 kilograms of fruit candy, and 150 kilograms of crunchy candy are needed.

In **Class 5F**, the ratio of **boys to girls** is **5:3**. It is known that there are **14 more boys than girls** . How many boys and girls are there in total?

$$5 - 3 = 2$$

$$14 \div 2 = 7$$



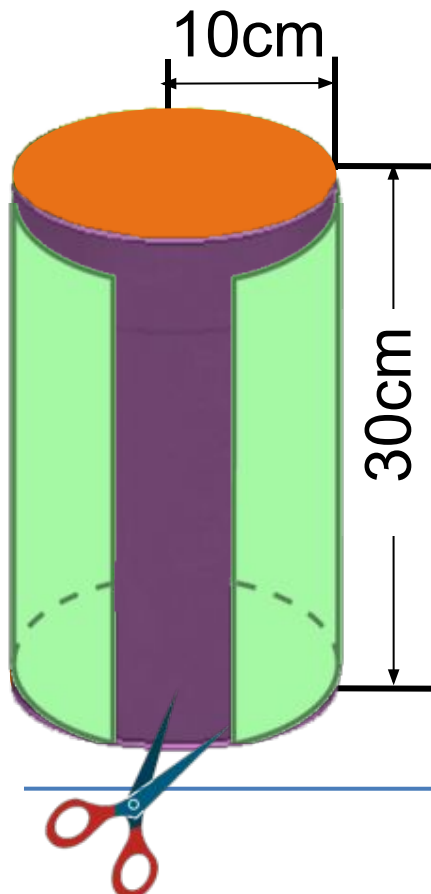
$$\text{boys: } 7 \times 5 = 35 \text{ (boys)}$$

$$\text{girls: } 7 \times 3 = 21 \text{ (girls)}$$

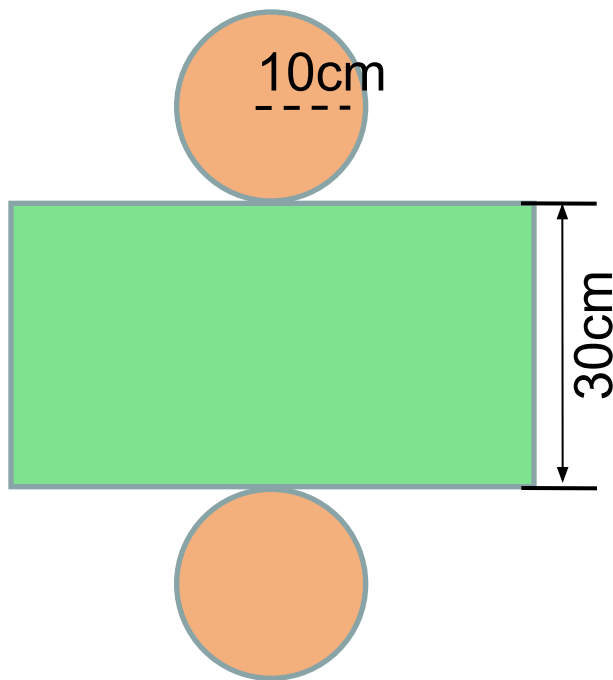
Cylinders and Cones

As shown in the picture, a cylindrical cardboard box is to be made. If the seam is not considered, what is the minimum area of cardboard required?

$$S_{side} = Ch$$



Explore



$S_{lateral}$:

$$2 \times 3.14 \times 10 \times 30 = 1884 (\text{cm}^2)$$

S_{base} :

$$3.14 \times 10^2 \times 2 = 628 (\text{cm}^2)$$

S_{total} :

$$1884 + 628 = 2512 (\text{cm}^2)$$

Answer: At least 2512 square centimeters of cardboard is needed.

The ground area occupied by a cylindrical pool. (**Base area**)

The area of iron sheet required to make a section of a chimney.
(**Lateral surface area**)

The area of the label paper on a can. (**Lateral surface area**)

The area of iron sheet required to make a tea canister.
(**Lateral surface area + 2 base area**)

The area of iron sheet required to make an open bucket.
(**Lateral surface area + 1 base area**)

The area rolled by the drum of a road roller in one rotation.
(**Lateral surface area**)

Conclusion

When solving problems related to "finding the surface area of a cylinder," it is important to clarify which part of the area is required in the problem. Generally, there are three cases:

- (1) Two bases and one lateral surface, such as cookie boxes, tea canisters, etc.
 - (2) One base and one lateral surface, such as open buckets, cylindrical fish tanks, etc.
 - (3) No bases, only the lateral surface is required, such as pipes, chimneys, road rollers, etc.
-

Discussion

⋮
Given the radius and height of a circle, how do you calculate the volume of a cylinder?

$$V = \pi r^2 h$$

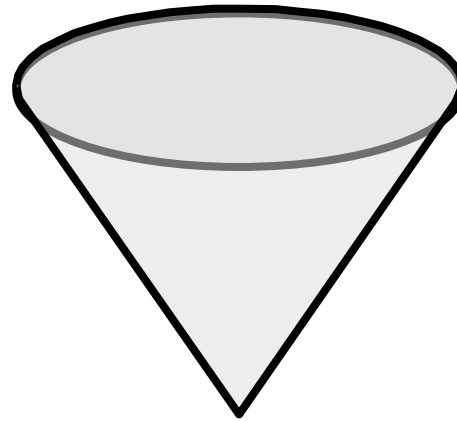
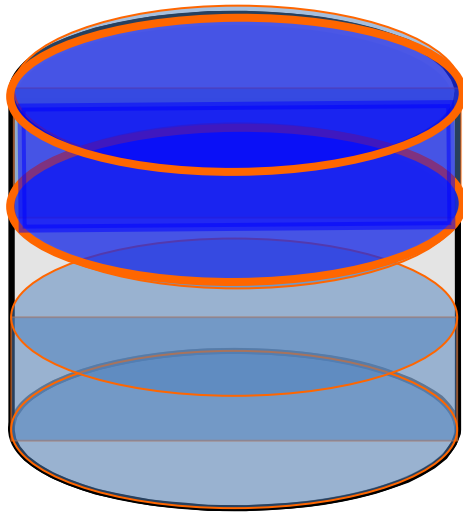
Given the diameter and height of a circle, how do you calculate the volume of a cylinder?

$$V = \pi (d \div 2)^2 h$$

Given the circumference and height of a circle, how do you calculate the volume of a cylinder?

$$V = \pi (C \div \pi \div 2)^2 h$$

What is the relationship between the volumes of a cylinder and a cone with the same base and height?



$$V_{cylinder} = \text{base area} \times \text{height}$$

$$V_{cone} = \text{base area} \times \text{height} \times \frac{1}{3}$$

Conclusion

For a cylinder and a cone with the same base and height, the volume of the cone is $\frac{1}{3}$ of the volume of the cylinder.

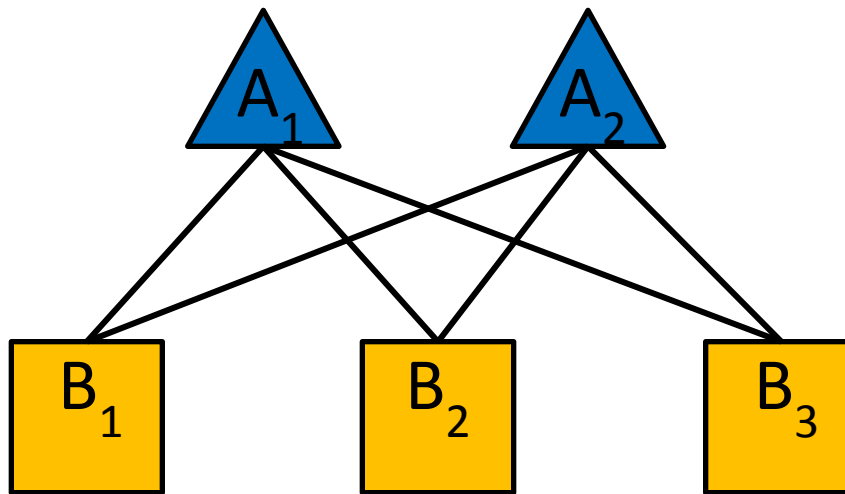
For a cylinder and a cone with different bases and height, the volume of the cone is **not** $\frac{1}{3}$ of the volume of the cylinder.

Permutation and Combination

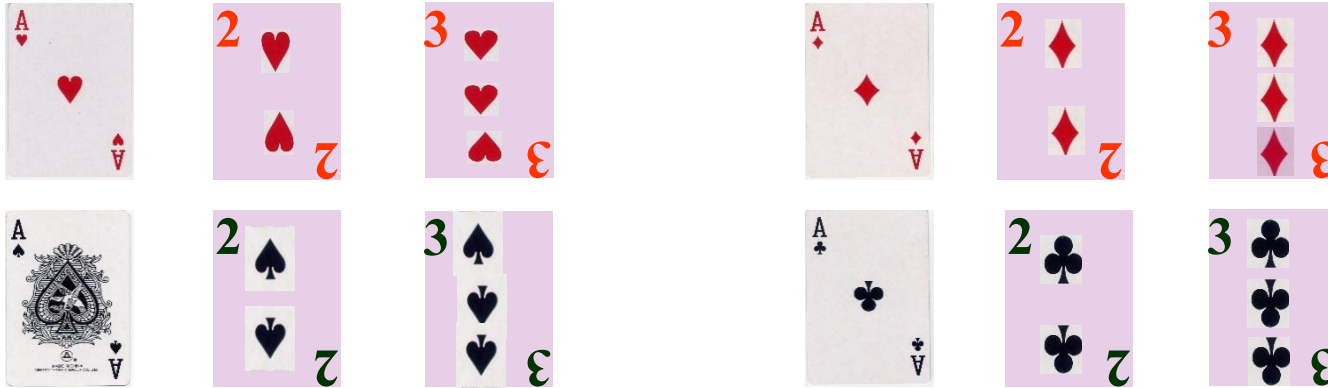
Introduction to Multiplication Principles

- Counting is the foundation of probability, statistics, and combinatorics.
- The **Multiplication Principle** is a key tool for solving counting problems. The clown in the circus is going to perform and wants to choose a hat and a pair of pants. How can they be matched?





Shuffle the following 12 face-down cards and draw one at random.



- ① The probability of drawing a heart is (), $\frac{1}{4}$
 The probability of drawing each of the other suits is (). $\frac{1}{4}$
- ② The probability of drawing an "A" is (), $\frac{1}{3}$ and the probability of drawing a 2 or 3 is () $\frac{1}{3}$
- ③ The probability of drawing the "Ace of Heart" () $\frac{1}{12}$
 which is (*different*) from the probability of drawing an "A".

Permutation

•**Definition:** A form of arrangement that choose r items from total of n items and then arranging them according to a specific requirement.

•**Notation:** n_{P_r} , $P(n, r)$, A_n^r

•**Formula:** $n_{P_r} = P(n, r) = \frac{n!}{(n-r)!}$

•**Example:** 5_{P_3} means 3 items chosen from 5 items for arrangement.

$$5_{P_3} = P(5, 3) = \frac{5!}{(5-3)!} = \frac{5!}{2!} = \frac{5 \times 4 \times 3 \times 2 \times 1}{2 \times 1} = 60$$

Combination

•**Definition:** A form of arrangement that choose r items from total of n items and then arranging them according to a specific requirement.

•**Notation:** n_{C_r} , $C(n, r)$, $\binom{n}{r}$, C_n^r

•**Formula:** $n_{C_r} = C(n, r) = \binom{n}{r} = C_n^r = \frac{n!}{r!(n-r)!}$

•**Example:** You have 5 fruits: 🍎, 🍌, 🍇, 🍊, 🍓 . How many ways can you choose 3 fruits?

$$\bullet 5_{C_3} = C(5, 3) = \binom{5}{3} = C_n^r = \frac{5!}{3!(5-3)!} = \frac{5!}{3!2!} = \frac{5!}{2!2!} = \frac{5 \times 4 \times 3 \times 2 \times 1}{3 \times 2 \times 1 \times 2 \times 1} = 10$$

Key Differences

Aspect	Permutation	Combination
Order	Matters	Doesn't matter
Formula		
Example	Arranging medals 🏆	Choosing a team 👥
Count	More outcomes	Fewer outcomes

Pigeonhole (Drawer's) Principle

A bag contains 10 red, 9 white, 8 yellow and 2 blue marbles. How many marbles must be taken out from the bag so that there will be 4 marbles of the same color ?

<u>R</u>	<u>W</u>	<u>Y</u>	<u>B</u>
1	1	1	1
1	1	1	1
1	1	1	

$$4 + 4 + 3 = 11$$

11 marbles have been taken out.

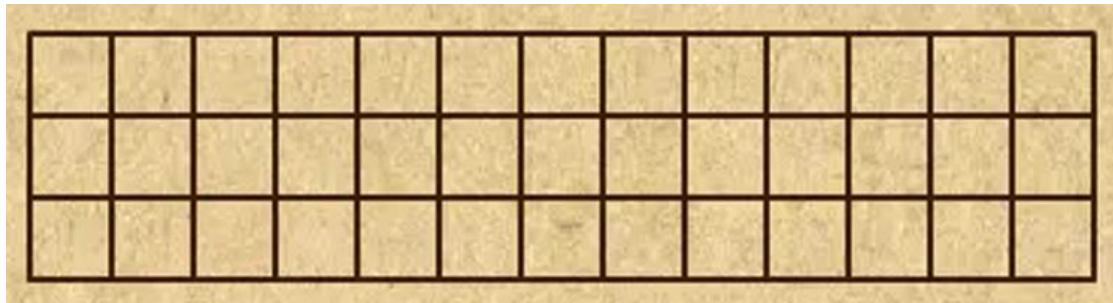
$$11 + 1 = 12$$

12 marbles must be taken out from the bag so that there will be 4 marbles of the same color.

In a table with 3 rows and 13 columns, using only black and white for coloring, will there be two columns with identical coloring patterns?

13 columns

3 rows



Multiplication Principle

$$2 \times 2 \times 2 = 8 \text{ (coloring patterns)}$$



8 drawers

$$13 \div 8 = 1 \dots 5$$

It is certain that 1+1=2 columns will have identical coloring patterns.

From the nine numbers 1 to 9, how many numbers must be selected to ensure that there are at least two numbers whose sum is 10?

1,2,3,4,5,6,7,8,9

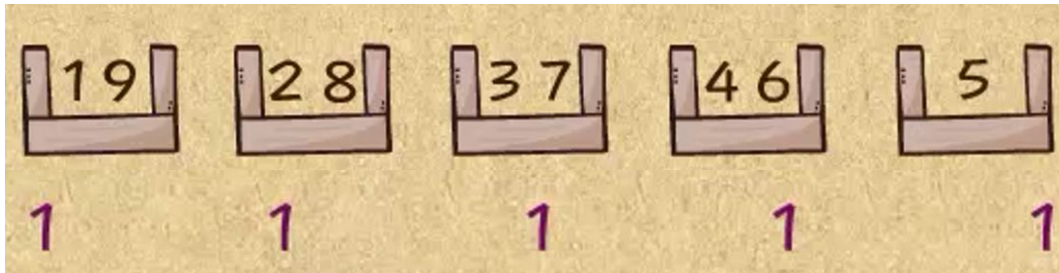
1+9

2+8

3+7

4+6

5

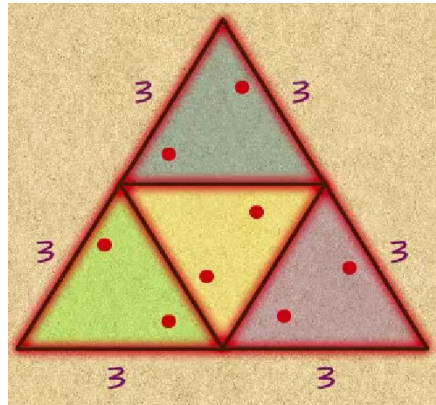
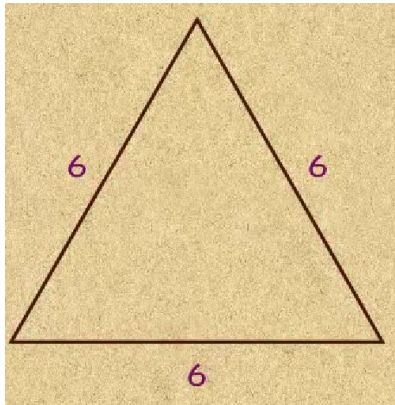


+1

$$5+1=6$$

At least 6 numbers must be selected to ensure that there are two numbers whose sum is 10.

In an equilateral triangle with a side length of 6 cm, if 5 points are placed randomly, is it certain that at least two points will be no more than 3 cm apart?



In each colored small triangle, any two selected points have a distance of no more than 3 cm.

$$5 \div 4 = 1 \dots 1$$

Each small triangle can be considered as a drawer.

There must be $1+1=2$ points in the same triangle, so these two points have a distance of no more than 3 cm.

